Social Science & Medicine 371 (2025) 117904

Contents lists available at ScienceDirect

Social Science & Medicine

journal homepage: www.elsevier.com/locate/socscimed

Bracing for impact: An intensive longitudinal investigation of weight stigma, vigilant coping, and maladaptive eating☆,☆☆

Karen E. Wetzel *, Mary S. Himmelstein, Jeffrey A. Ciesla

Department of Psychological Sciences, Kent State University, United States

ARTICLE INFO

Handling editor: Cecilia Cheng

Keywords:
Weight-based discrimination
Vigilance
Eating
Ecological momentary assessment

ABSTRACT

Objective: Interpersonal weight stigma (being teased, treated unfairly, or discriminated against for body weight) increases one's stress and negatively impacts their well-being. Weight stigma consistently increases maladaptive eating behaviors, such as eating to cope, binge eating, and restrictive dieting. Previously published cross-sectional models suggest that vigilant coping (being on the lookout for future discrimination) is one way in which weight stigma may impact maladaptive eating behaviors, especially as internalized weight stigma (self-stigma) increases. This study aimed to analyze these relationships longitudinally using ecological momentary assessments.

Methods: Female, higher weight college students ($M_{BMI} = 32.7$, $SD_{BMI} = 6.09$) from a large public midwestern university ($N = 130$) completed 3 surveys per day for 7 days, reporting their vigilant coping, eating to cope, restrictive dieting, and binge eating. Data was collected from January 2023–February 2024. We examined the relationship between baseline experienced and internalized weight stigma and day-to-day vigilant coping. Additionally, we examined vigilant coping earlier in the day as a predictor of eating later in the day.

Results: Experienced and internalized weight stigma did not interact to predict day-to-day vigilant coping, but they both independently predicted vigilant coping. Vigilant coping earlier in the day predicted restrictive dieting later in the day but did not predict emotional or binge eating. There were no significant conditional indirect effects.

Conclusions: This study establishes vigilant coping as a longitudinal predictor of restrictive dieting, but further research is required to understand other health behaviors and outcomes related to vigilant coping.

A growing body of research revolves around weight stigma, social devaluation due to body weight (Hunger et al., 2015). This area of research is especially timely given increases in the prevalence of higher body weights (here, BMI $\geq$30 kg/m^2; Hales, 2020) and the rise in negative attitudes toward people with higher body weights; Charlesworth and Banaji (2019). People with higher body weights experience hurtful and pervasive stigma, occurring at both the interpersonal and systemic levels (Gerend et al., 2021). These weight stigmatizing experiences, in turn, impact people's health and well-being (Puhl et al., 2020; Vartanian and Porter, 2016).

Experienced weight stigma is theorized to impact people via internalized weight stigma (IWS), or self-stigmatization due to one's body weight (Durso and Latner, 2008; Pearl and Puhl, 2014). Both experienced stigma and IWS are uniquely related to poorer physiological & psychological health (Wu and Berry, 2018), and IWS is associated with finding stigmatizing experiences distressing (Pudney et al., 2020). IWS is difficult to change over time (Pearl et al., 2023), and thus may be better suited as a moderator of the effects of weight stigma.

1. Outcomes of weight stigma

Ultimately, weight stigma is stressful and harms individuals' health. Experienced weight stigma is related to increased depression and anxiety (Sutin and Terracciano, 2017; Wellman et al., 2019), increased avoidance of physical activity and lower motivation to exercise (Carels et al., 2019; Pearl et al., 2021), and increased substance use (Gerend et al., 2023).

Perhaps the most widely researched outcome of experienced weight

☆ We have no conflicts of interest to disclose. Pre-registration, data, analyses, and measures can be viewed on Open Science Framework (https://osf.io/sz38g/).☆☆ This project was funded by Kent State University's Graduate Student Senate and the Judie Lasser Award.

* Corresponding author. Department of Psychological Sciences, Kent State University, PO Box 5190, Kent, OH, 44242, United States.
E-mail address: karenewetzel@gmail.com (K.E. Wetzel).

https://doi.org/10.1016/j.socscimed.2025.117904
Received 10 September 2024; Received in revised form 12 January 2025; Accepted 28 February 2025
Available online 28 February 2025
0277-9536/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

stigma is maladaptive eating (for review, see: Levinson et al., 2024; Vartanian and Porter, 2016), largely due to weight stigma's consistent relationship with maladaptive eating outcomes. These eating outcomes are also commonly studied because they link weight stigma with future weight gain (Puhl et al., 2020; Tomiyama, 2014).

Most work establishing weight stigma as a predictor of eating outcomes is cross-sectional in nature and substantiates a relationship between weight stigma and binge eating (Levinson et al., 2024; Vartanian and Porter, 2016). Experimental work on eating outcomes of weight stigma shows that viewing weight stigmatizing content is associated with eating more high calorie foods (Major et al., 2014).

2. Vigilant coping

Social identity threat is the concern that occurs when one believes they may be stigmatized because of group membership, and it is one proposed way that stigmatization impacts one's health (Crocker and Major, 1989; Major and O'Brien, 2005; Steele et al., 2002). Social identity threat theory originated from research on social categorization and identity theory, which explores how we define and judge groups (Tajfel, 1970; Tajfel et al., 1971), as well as stereotype threat theory, which found individuals experience a threat response when they are worried about confirming negative stereotypes about their group (Spencer et al., 1999; Steele et al., 2002; Steele and Aronson, 1995). Early work on social identity threat suggests that even in the absence of experienced stigma, perceiving threat of stigmatization can impact one's well-being (Major and O'Brien, 2005). Because threat can occur even in the absence of danger, people may continuously experience stress, which is physiologically dysregulating (Hunger et al., 2015). A growing literature applies social identity threat to weight stigma (Major et al., 2014; Hunger et al., 2015; Major and Schmader, 2018). It is important to note that one does not need to self-categorize as high weight to experience weight-based social identity threat; knowing others may categorize them as high weight and devalue them is enough to induce weight-based social-identity threat (Hunger et al., 2015)."

Social identity threat has recently been extended to the weight stigma literature, where it has been established as a mechanism of weight stigma and health (Hunger et al., 2018, 2020). In the lab, weight salience increases the expectation that one will be rejected for their weight among women (Blodorn et al., 2016), and weight stigmatizing content increases concerns about becoming targets of weight stigma among women with high weight (Major et al., 2014). A commonly examined threat response in the racism literature is vigilant coping or being 'on the lookout' for discrimination (Clark et al., 1999; Sawyer et al., 2012). When people think they may be stigmatized, they attempt to detect and avoid stigmatization via hypervigilance and behavior monitoring (Sewell et al., 2016; Steele et al., 2002). The literature on racism suggests that this vigilant coping can be passed down from family (Brownlow, 2023; Sewell et al., 2016) and may be adaptive in many ways (i.e., short-term physical safety). However, over the long-term it has negative consequences for people's health and well-being (Hicken et al., 2018; Himmelstein et al., 2015).

Weight-related vigilant coping has seldom been studied. A model of vigilant coping has been cross-sectionally established as a mediator of experienced weight stigma and eating outcomes (Wetzel and Himmelstein, 2023a,b). In experimental paradigms, both weight- and race-related vigilant coping have been measured by proxy of perceiving more stigmatization in an ambiguous interaction and heightened physiological responses (Hunger et al., 2018; Sawyer et al., 2012).

3. Ecological momentary assessment (EMA)

There is limited work on weight stigma using EMA studies, which are composed of surveys that are administered throughout the day to be able to capture real-time data (Bolger and Laurenceau, 2013), most of which is exploratory (for review, see Bidstrup et al., 2023). A recent review by Bidstrup and colleagues (2023) revealed that much of the literature is currently limited to descriptive statistics rather than inferential. At the time of that review, only 12 studies had assessed weight stigma using EMAs. Overall, studies of this nature find that weight stigmatizing experiences are low frequency but impactful (Potter et al., 2021), suggesting that a more frequent stressor such as vigilant coping may be worth examining. Some EMA studies have instead looked at weight-related stressors, rather than explicit weight stigma (e.g., instances of shame or stress around one's weight), and found that these experiences are more common (Carels et al., 2019).

EMA research on eating outcomes of weight stigma is limited, and most are concerned only with between-subjects processes (associations between variables measured from different people and influenced by individual differences; e.g., comparing eating patterns of people have experienced weight stigma relative to those who have not; Bidstrup et al., 2023). This is likely due to the infrequency of experienced weight stigma itself, which makes it difficult to look at effects within-person (relationship between two variables measured from the same person across time, for example examining whether a person who experienced weight stigma early in the day engages in disordered eating later in the day; Potter et al., 2021) and no proposed mediators being tested (Bidstrup et al., 2023). Currently, severe limitations exist in this line of research with very few pre-registered studies, and nearly all work having small sample sizes (Bidstrup et al., 2023).

While still relying on recall and often self-report, EMA studies are meant to capture short-term behaviors and in-the-moment emotions. Thus, EMA is ideal for research questions involving the temporal precedence of within-day effects (factors that may impact each other within the same day). Additionally, long-term longitudinal studies often are not ideal for eating behavior research due to the long-term recall required and people's inabilities to accurately recall behavior over such an interval, versus the short-term recall required for EMA studies (Smyth et al., 2001).

4. Current study

This study will examine a model of weight stigma and eating ((Wetzel and Himmelstein, 2023a,b; See Figs. 1 and 2) using EMA. First, we hypothesize that experienced weight stigma (EWS) will be positively related to vigilant coping, and this relationship will become more positive as IWS increases (moderation effect). Additionally, we hypothesize that vigilant coping at one timepoint will positively predict maladaptive eating at the following timepoint. Finally, we hypothesize that there will be a conditional indirect effect between weight stigma and eating outcomes via vigilant coping, such that the indirect effect becomes more positive as IWS increases (See Fig. 1). The use of EMA in this study is ideal to establish the temporal precedence of a model that has been supported by multiple cross-sectional samples but requires support for effects over time. This is also the best way to acquire data about one's day-to-day experiences and temporal precedence of stigma, vigilance, and eating behaviors in one's natural environment.

5. Methods

Participants (N = 130) were recruited from a large midwestern university subject pool from January 2023 through February 2024. Individuals who completed an optional prescreen and indicated they identified as female, perceived themselves as higher weight, and had never received a formal eating disorder diagnosis were eligible to sign up for the baseline lab visit. The study advertisement on the subject pool website noted that participants may be eligible to complete a week-long study for monetary compensation via Amazon gift cards based on their initial lab visit. The study was approved by the institutional IRB (Protocol #322) in October of 2022. The study was originally pre-registered at https://doi.org/10.17605/OSF.IO/PF9CU with substantial edit made in February 2024, after the author had completed sufficient statistical

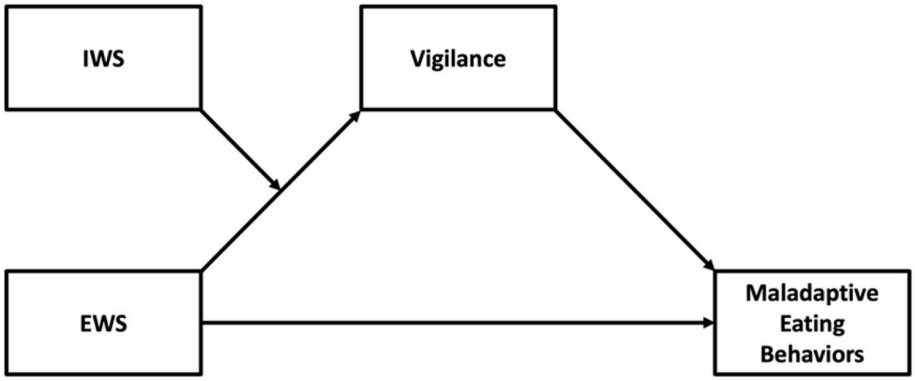

Fig. 1. Theoretical model of primary hypothesis.
Note. EWS = experienced weight stigma, IWS = internalized weight stigma.
Caption: The figure includes a moderated mediation model in which EWS points to Vigilance and Maladaptive Eating Behaviors. IWS points to the arrow from EWS to Vigilance. Vigilance also points to Maladaptive Eating Behaviors.

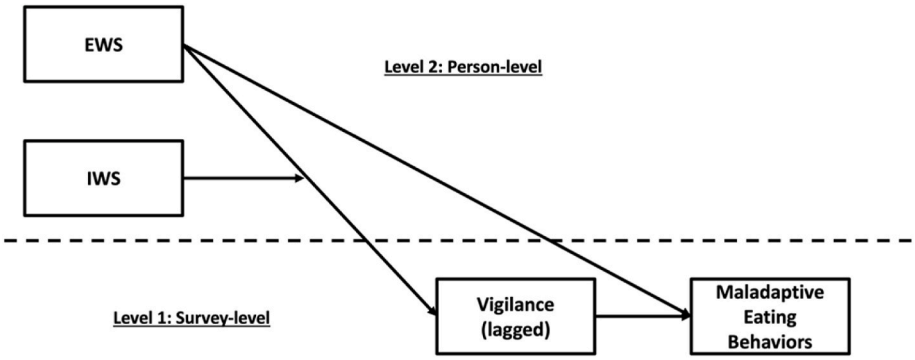

Fig. 2. Multilevel translation of theoretical model.
Note. EWS = experienced weight stigma, IWS = internalized weight stigma; Only focal variables included in model. Other effects were included in the analyses that are not modeled here for interpretability.
Caption: The figure includes two levels – Level 2: Person-level and Level 1: Survey-level with a dashed line between the two. EWS and IWS are both in the person-level section. Vigilance (lagged) and Maladaptive Eating Behaviors are in the survey-level section. EWS points to Vigilance (lagged) and Maladaptive Eating Behaviors, both crossing from person-level into survey-level. IWS points to the arrow from EWS to Vigilance (lagged). Vigilance (lagged) points to Maladaptive Eating Behaviors, staying in the survey level.

training to more adequately describe the analytic plan before data were analyzed. Deviations from pre-registration and updates are noted in the methods. An a priori power analysis was done using G*power, estimating sample size for repeated measures ANOVA as a floor for the sample of N = 96 with 21 surveys (F = 0.1 at 0.95 power; 3 groups, 21 measures). The plan was to sample at least 120 participants with at least 5 surveys to ensure power for analyses.

5.1. Participants

A total of 198 participants arrived to complete the baseline lab visit. Two participants were ineligible due to previous completion of the study and were granted credit for the visit but not permitted to complete the momentary assessments again and were not reassessed. An additional 40 participants were deemed ineligible due to the eligibility criteria.

This study was limited to cisgender women due to constraints of resources and their unique relationships to weight-related stressors. In experimental paradigms, women who perceive themselves as high body weight are most stressed by weight stigma and threatened by the risk of weight stigma (Himmelstein et al., 2015; Major et al., 2014). Additionally, due to the limited number of higher weight men in the subject pool, we would be unable to recruit enough men with higher body weights to examine gender differences in these relationships.

5.2. Baseline lab visit protocol

Inclusion criteria for the week-long study were determined during the baseline lab visit. If the participant met inclusion criteria, the research assistant allowed them to complete the full baseline survey, assisted in downloading the SEMA3 app, gave them a handout for using the app, and measured their height and weight using a WB-3000 Digital Physician's Model scale and stadiometer.

Height and weight were assessed after completing the baseline surveys to lower the chances of this impacting due to body salience or social identity threat. To lessen the impact of weight measurement on participants, they were not informed of their weight unless they specifically asked for it (i.e., a card was placed over the scale monitor while individuals were weighed), research assistants were trained on how to measure weight respectfully, and there were mental health handouts in case a participant reported distress. Specifically, research assistants were taught to remain professional throughout the interaction, maintain a neutral facial expression when recording a participant's weight, refrain from making any comments about weight, and offer to take self-reported weight if a participant seemed hesitant about being weighed. All included participants consented to having their weight measured in the lab. Additionally, participants received local mental health resources and were informed that they could end participation at any time.

5.3. Momentary assessment protocol

Eligible participants completed momentary assessments via the SEMA3 app. Participants completed three signal-contingent surveys per day over the course of a standard week, resulting in 21 total surveys. The first survey each day was completed within an hour of waking. They were also prompted once in the mid-afternoon at a semi-random time and once at night, which they were instructed to complete within an hour of going to bed.

5.4. Compensation

Participants were provided course credit for their baseline lab visit. Participants who were able to download the app and complete daily surveys were compensated $0.50 per survey and entered once into a raffle for six, $70 Amazon gift cards per completed survey. The gift cards were raffled at the end of data collection.

5.5. Measures

5.5.1. Baseline

Gender. To determine eligibility, participants were asked, "How would you describe your gender?" with the options of "*Female*," "*Male*," "*Non-binary*," or "*Other*". Participants had to indicate "Female" to be eligible for the 7-day EMA study. The question in the prescreen which allowed participants to sign up for the baseline assessment of eligibility asked, "How do you identify" with the following options "Male/Man," Female/Woman," "Transgender Man," "Transgender Woman," "Non-binary," "Something not listed," and "Prefer not to say." Participants had to select "Female/Woman" to be able to sign up for the in-lab assessment.

Eating Disorder Diagnosis. As part of eligibility, participants were asked, "Have you been diagnosed by a physician, psychologist, or other clinician with an eating disorder?" with the options of "*Yes*" or "*No*." Participants had to indicate "No" to be eligible for the 7-day EMA study.

Mental Health Hospitalization. Again, to assess eligibility participants were asked, "Have you been hospitalized within the past month for your mental health?" with the options of "*Yes*" or "*No*." Participants had to indicate "No" to be eligible for the 7-day EMA study.

Perceived Weight. Participants were asked how they would describe their weight with the options of "*Underweight*," "*Average weight*," "*Overweight/a little heavier than others*," or "*Obesity/much heavier than others*" (Himmelstein and Tomiyama, 2015; Robinson et al., 2015). Those who selected one of the two higher weight categories could be eligible.

Access and Willingness for EMA. Finally, participants were asked questions about their access to a smartphone and willingness to complete daily surveys, to which they could respond "Yes" or "No". The questions were: (1) "Do you have a smartphone available throughout the day that you can and are willing to download a survey administration app to?", (2) "Would you be willing to download an app to complete a one-week study?", and (3) "Would you be willing to complete 3 short surveys (2–5 min each) per day for a total of one week?"

Weight Stigma. Three items assessed lifetime experience with interpersonal weight stigma. Participants were asked "Have you ever been [teased/treated unfairly/discriminated against] for your weight?" (Puhl et al., 2011). Response options to each question were 0 (*No*) and 1 (*Yes*). Following previous research, responses were recoded into 0 = "No" to all items and 1 = "Yes" to any of the three items.

Internalized Weight Stigma. The Weight Bias Internalization Scale-Modified (WBIS-M) was used to measure internalized weight stigma, α = .93 (Durso and Latner, 2008; Pearl and Puhl, 2014). The scale includes 10 items, such as, "I don't feel that I deserve to have a really fulfilling social life, because of my weight," and responses range from (1) *Strongly Disagree* to (7) *Strongly Agree.* The mean of the 10 items is used as each person's score, $M = 4.60$, $SD = 1.40$.

BMI. Body mass index in kg/m^2 was calculated using height and weight taken in lab. Participants were not required to be in a higher BMI category (BMI >25 kg/m^2) since perceptions of weight typically drive social identity threat mechanisms (Hunger et al., 2015; Major et al., 2014). BMI was selected as a covariate for analyses to separate the impact of weight stigma and weight-related stress from the overall impact of body mass on the relationship between stigma and eating outcomes. For demographic information, see Table 1.

Race. Race was assessed with one question: "How would you classify yourself?" with the options of "White, non-Hispanic, non-Latino," "Black or African American," "Black, Hispanic/Latino," "Asian or Pacific Islander," "Latino, Hispanic, or Mexican American," "Middle Eastern/North African," "Indigenous or Native American," "Biracial or Multiracial," or "Something else" with an option to describe if selecting the final option. Because of limited diversity in the sample and statistical power, this was dummy coded into *0 = White, non-Hispanic* and *1 = Any other racial or ethnic identity* for covarying race/ethnicity in the models.

5.5.2. Momentary assessments

Weight Stigma. Weight stigma was assessed at each time point using the same three questions as at baseline as well as agreement with the item, "I encountered a situation in which something was physically too small or weight limit too low for comfortable participation (i.e., classroom desk or chair, bus seat, a roller coaster, gym equipment, blood pressure cuff)" to assess physical sources of discrimination. They responded from (1) *Strongly Disagree* to (7) *Strongly Agree,* and these responses are recoded to 1–4 = 0 "No", 5–7 = 1 "Yes", $M = 0.03$, $SD = 0.08$.

Vigilance. Vigilance was assessed at each time point using 4 items from the Williams Heightened Vigilance Scale (WHVS; Brownlow et al., 2023; Himmelstein et al., 2015; Jackson and Williams, 2002; Wetzel and Himmelstein, 2023). An example item is "Since the [last survey], I: Tried to avoid certain social situations and places because of my weight." The scale was rated (1) *Strongly Disagree* to (7) *Strongly Agree*, $M = 2.49$, $SD = 1.14$, α = .90-0.92.

Eating to Cope. Eating to cope was measured using two items from the Emotional Eating subscale of the Motivations to Eat Scale (e.g., "Since the [last survey], I [ate as a way to help me cope," B. Jackson et al., 2003). These items were scored (1) *Strongly Disagree* to (7) *Strongly Agree* and averaged to determine eating to cope at that time point, $M = 2.20$, $SD = 0.91$, α = .83-0.89.

Binge Eating. Overeating with a loss of control was measured using three items from the EDE-Q (Fairburn and Beglin, 1994). Participants were asked: "Since [the last survey] I ate what other people would regard as an unusually large amount of food (given the circumstances) at

Table 1
Demographic information.

	Sample ($N = 130$)	
	M	*SD*
Age	20.6	5.08
BMI	32.7	6.09
	N	%
Race/Ethnicity		
Asian or Pacific Islander	8	6.2
Black or African American	21	16.2
Black, Hispanic/Latino	2	1.5
Latino, Hispanic, or Mexican American	3	2.3
Biracial or Multiracial	5	3.8
White, non-Hispanic or Latino	87	66.9
Something Else/Not Listed	4	3.1
BMI Category		
<18.5	0	0
18.5 to 24.9	8	6.2
25–29.9	40	30.8
≥ 30	82	63.1

least once." If they responded at least "Somewhat Agree," they were then asked to rate the response, "I had a sense of having lost control over my eating (at the time I was eating)." Participants were asked to rate these items from (1) *Strongly Disagree* to (7) *Strongly Agree.* Then, participants were asked, "How many times did this occur since [the last survey] (overeating with a sense of having lost control)?" and allowed to rate it as (1) *Once,* (2) *Twice,* (3) *Three or more times*. These items were recoded so that 0 = no overeating and no loss of control, 1 = overeating but no loss of control, 2 = overeating with loss of control once, 3 = overeating with loss of control twice, $M = 0.18$, $SD = 0.27$.

Restrictive Eating. To assess restraint in eating, participants were asked to rate two items from the EDE-Q 9 (Fairburn and Beglin, 1994): "I deliberately tried to limit the amount of food I eat to influence my shape or weight since [the last survey]," and "I tried to follow definite rules regarding my eating (for example, a calorie limit) in order to influence my shape or weight since [the last survey]" from (1) *Strongly Disagree* to (7) *Strongly Agree.* These were averaged to determine a score for restrictive eating at that timepoint, $M = 2.90$, $SD = 1.32$, $\alpha = .87\text{-}0.89$.

5.6. Data analytic plan

Data were analyzed using SPSS Version 29.0, using the mixed command to test mixed-effects models. For testing the proposed model (See Fig. 2), the MLmed macro was used (Rockwood, 2017). This macro allows 2-1-1 moderated mediation analyses, as tested here, in accordance with recommendations from scholars in avoiding the ecological fallacy (e.g., the issue of predicting a level-1 outcome with a level-2 predictor; See Fig. 3; Preacher et al., 2010; Robinson, 1950; Rockwood, 2017; Selvin, 1958; Zhang et al., 2009). All models were run as restricted maximum likelihood to enhance power. All results were corrected for family-wise error using Benjamini & Hochberg's recommendations (Benjamini and Hochberg, 1995). This correction did not impact interpretation of any results. There were no practically significant univariate outliers in the data (except for time, which is previously addressed).

To test the data's fit for multilevel analyses, an "empty model" was run, including only the nesting variable (participant) to predict the outcomes (three eating outcomes). Acceptability of multilevel structure was determined by a significant random effect of person on each eating outcome. This substantiates that there was sufficient variability between different people to be considered multilevel. Simply, because the data is naturally nested given the EMA design (7 surveys per person), multilevel data analysis is necessary and expertly handles random missing data between people, and our data was determined to be fit for multilevel analyses.

Path A of the model (see Fig. 3) was analyzed – see Eqn (1 and 2) – with Level 2 focal variables of baseline weight stigma, internalized weight stigma, and their interaction term predicting the random intercept of vigilant coping. This model compares vigilance between those who have experienced stigma to those who have not and whether this is influenced by IWS.

$$VIG_{ij} = \pi_{0j} + e_{ij} \tag{1}$$

$$\pi_{0j} = \beta_{00} + \beta_{01}(BMI - \overline{BMI}) + \beta_{02}\,(RACE) + \beta_3(EWS) + \beta_{04}(IWS - \overline{IWS}) + \beta_{05}(EWS^*[IWS - \overline{IWS}]) + r_{ij}) \tag{2}$$

Then, path B was analyzed within-person with lagged, person-mean centered vigilance on eating, as seen in Eqn (3, 4, 5 and 6). This specifies that the eating outcome was predicted by vigilance at the previous timepoint, within-person. The random effect was included for Eqn (5 and 6) only when chi-square analysis of nested model testing was significant. This model tests whether maladaptive eating (i.e., restrictive dieting, eating to cope, binge eating) occurs after experiencing more vigilance than one typically does.

$$EAT_{ij} = \pi_{0j} + \pi_{1j}\,(time) + \pi_{2j}\,\left(VIG_{ij} - VIG_{\cdot j}\right)_{t-1} + \pi_{3j}\,\left(EAT_{ij}\right)_{t-1} + e_{ij} \tag{3}$$

$$\pi_{0j} = \beta_{00} + \beta_{01}(BMI - \overline{BMI}) + \beta_{02}(race) + \beta_{03}(EWS) + \beta_{04}\left(VIG_{\cdot j}\right) + r_{ij} \tag{4}$$

$$\pi_{1j} = \beta_{10} + r_{1j} \tag{5}$$

$$\pi_{3j} = \beta_{30} + r_{3j} \tag{6}$$

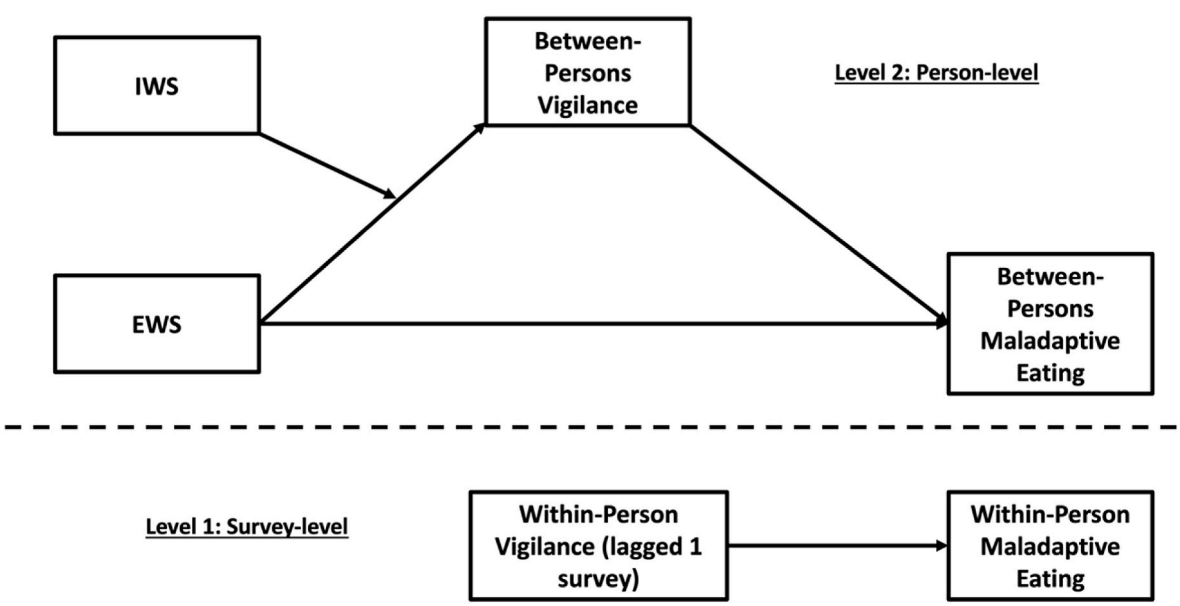

Fig. 3. Multilevel model as analyzed in accordance with Preacher et al., 2010 recommendations for 2-1-1 mediation.
Note. IWS = internalized weight stigma, EWS = experienced weight stigma.
Caption: The figure includes two levels – Level 2: Person-level and Level 1: Survey-level with a dashed line between the two. EWS, IWS, Between-persons vigilance, and between-persons maladaptive eating are all in the person-level section. Within-person vigilance (lagged 1 survey) and Within-person Maladaptive Eating are in the survey-level section. EWS points to Between-persons vigilance and Between-persons Maladaptive Eating. IWS points to the arrow from EWS to Between-persons Vigilance. Within-person Vigilance (lagged 1 survey) points to Within-person Maladaptive Eating. All relationships stay within level.

6. Results

Participants had an average age of 21 years old and an average BMI of 32.7 kg/m^2 6.2% of participants had a BMI in the "normal" weight range, and the rest were either in the "overweight" or "obese" BMI range. 67% of participants were White and non-Hispanic, 16% were Black and non-Hispanic, and 6% identified as Asian or Pacific Islander. BMI ranged from 20.2 kg/m^2 to 53.0 kg/m^2, M = 32.7 kg/m^2. At baseline, 83.8% of participants reported having experienced interpersonal weight stigma (teasing, unfair treatment, or discrimination) at least once in their lifetime. 78.5% of participants (N = 102) reported weight teasing, 53.1% (N = 69) reported unfair treatment due to weight, and 41.5% reported weight discrimination at baseline. For correlations of experienced weight stigma, internalized weight stigma, and BMI, see Table 2.

There was a response rate of 67.8% among included participants, resulting in 1754 total surveys submitted. After exclusion for data quality concerns, there were 1502 remaining surveys (58.6% of scheduled surveys completed). Listwise, there were 1089 timepoints included.

Throughout the week, there were 23 total timepoints where interpersonal weight stigma was reported (about 0.8% of responses), suggesting that it occurs infrequently. However, these 23 experiences were reported by 17 different participants, suggesting that many participants (13.1%) experienced weight stigma at least once. Instances of weight stigma, regardless of timepoint are as follows: weight teasing (N = 14), being treated unfairly (N = 10), weight discrimination (N = 9), encountering a space that was too small (N = 38).

A total of 13% (N = 196) of submitted momentary surveys reported at least some vigilant coping (M > 4.0) on average. However, some vigilance was reported at least once by 55 different participants (42% of the sample), suggesting that many people experience it, though typically only once every few days.

The empty models showed that the nesting structure was significant for all three eating outcomes, and between-person variance accounted for about 57% of the variance in restrictive dieting, 15% of the variance in binge eating, and 44% of the variance in eating to cope.

The trend in vigilant coping was analyzed across time, due to concerns that vigilance may increase due to the study. However, there was a small negative effect of time on vigilance, such that people reported less vigilance as time went on, $B = -0.01$, $p = .041$, 95% CI [−0.02, 0.00].

6.1. Focal analyses

Controlling for BMI and race, the interaction between experienced and internalized weight stigma ('a' path) did not predict vigilant coping, $B = 0.19$, $p = .258$, 95% CI [−0.14, 0.51]. Deviating from preregistration, we explored whether experienced and internalized weight stigma predicted vigilant coping without the interaction term in the model (See Table 3). Experienced weight stigma, $B = 0.60$, $p = .020$, 95% CI [0.10, 1.10], and internalized weight stigma, $B = 0.40$, $p < .001$, 95% CI [0.27, 0.52] predicted vigilant coping, such that greater internalized weight stigma and having experienced weight stigma both predicted greater day-to-day vigilant coping.

6.1.1. Restrictive dieting

Within-person vigilant coping positively predicted restrictive dieting at the next timepoint, $B = 0.20$, $p < .001$, 95% CI [0.10, 0.30] (See Table 4). Between-persons vigilant coping also positively predicted restrictive dieting, suggesting that people with higher vigilant coping also report higher restrictive dieting, $B = 0.69$, $p < .001$, 95% CI [0.49, 0.89].

Table 2
Correlations.

		1	2
1. BMI	32.68 (6.1)		
2. Experienced Weight Stigma	0.84 (0.37)	0.022	
3. Internalized Weight Stigma	4.60 (1.40)	0.115	0.287***

***. Correlation is significant at the 0.001 level (2-tailed).

Table 3
Between-person prediction of the intercept of EMA vigilant coping.

	B	SE	p	LLCI	ULCI
Model with Interaction					
Within					
Intercept	**1.82*** **	0.27	<0.001	1.30	2.35
Between					
BMI (centered)	0.01	0.01	0.388	−0.02	0.04
Not White (ref. White)	0.03	0.18	0.880	−0.33	0.39
EWS	**0.72***	0.27	0.010	0.18	1.27
IWS	0.24	0.15	0.111	−0.06	0.53
IWS * EWS	0.19	0.16	0.258	−0.14	0.51
Random					
Intercept	**0.79*** **	0.11	<0.001	0.60	1.05
Model for Main Effects					
Within					
Intercept	**1.93*** **	0.24	<0.001	1.44	2.41
Between					
BMI (centered)	0.01	0.02	0.459	−0.02	0.04
Not White (ref. White)	0.08	0.18	0.666	−0.28	0.44
EWS	**0.60***	0.25	0.020	0.10	1.10
IWS	**0.40*** **	0.06	<0.001	0.27	0.52
Random					
Intercept	**0.81*** **	0.12	<0.001	0.61	1.07

Note. BMI = body mass index in kg/m^2, EWS = experienced weight stigma, IWS = internalized weight stigma; Significant findings are bolded.

6.1.2. Binge eating

Vigilant coping at the previous timepoint did not predict binge eating at the following timepoint, $B = -0.04$, $p = .058$, 95% CI [−0.09, 0.001] (See Table 4). Between-persons vigilant coping predicted binge eating, $B = 0.05$, $p = .032$, 95% CI [0.004, 0.10], such that those who are more vigilant day-to-day also report more binge eating day-to-day.

Because of the distribution of the binge eating variable, Poisson and negative binomial models were tested using STATA 18.0/SE to confirm results (See Table 5).

6.1.3. Eating to cope

Vigilant coping at the previous timepoint did not predict eating to cope at the following time point, $B = 0.01$, $p = .790$, 95% CI [−0.08, 0.11] (See Table 4). Between-persons vigilant coping positively predicted the eating to cope, suggesting that people with higher vigilant coping also report greater eating to cope, $B = 0.40$, $p < .001$, 95% CI [0.26, 0.55].

6.1.4. Moderated mediation

Between-persons indices of moderated mediation were nonsignificant for all eating outcomes: restrictive dieting, IMM = 0.13, $p = .131$, eating to cope, IMM = 0.08, $p = .133$, binge eating, IMM = −0.03, $p = .314$.

The between-within indices of moderated mediation were also nonsignificant for all eating outcomes: restrictive dieting, IMM = 0.04, $p = .137$; binge eating, IMM = −0.01, $p = .329$; eating to cope, IMM = 0.03, $p = .396$.

7. Discussion

Overall, interpersonal weight stigma was infrequent (present in 0.8% of surveys), but vigilant coping was more common (present in 13% of surveys). Thus, while not measured in the same way, vigilance was reported approximately 8.5 times as frequently as experienced weight stigma, suggesting that vigilant coping is likely a more prominent day-to-day stressor for women with high body weights. However, contrary

Table 4
Mixed model of vigilant coping on eating outcomes.

	B	SE	p	LLCI	ULCI
DV: Restrictive dieting					
Within					
Intercept	**1.15*** **	0.30	<0.001	0.56	1.74
Vigilance $_{t-1}$[a]	**0.20*** **	0.05	<0.001	0.10	0.30
Time	−0.01	0.01	0.306	−0.02	0.01
Restrictive dieting $_{t-1}$	−0.06	0.04	0.165	−0.14	0.02
Between					
EWS	−0.30	0.32	0.360	−0.93	0.34
Vigilance[b]	**0.69*** **	0.10	<0.001	0.49	0.89
BMI[c]	**−0.04***	0.02	0.024	−0.08	−0.01
Not White (ref. White)	−0.05	0.23	0.837	−0.50	0.41
Random Effects					
RD Intercept	**1.27*** **	0.19	<0.001	0.95	1.69
Slope RD $_{t-1}$ → RD$_t$	0.03	0.02	0.072	0.01	0.08
DV: Binge Eating					
Within					
Intercept	−0.001	0.08	0.995	−0.15	0.15
Vigilance $_{t-1}$[a]	−0.04	0.02	0.058	−0.09	0.001
Time	0.005	0.003	0.067	−0.00	0.01
Binge Eating $_{t-1}$	**−0.10***	0.05	0.025	−0.19	−0.01
Between					
EWS	0.06	0.07	0.387	−0.08	0.20
Vigilance[b]	**0.05***	0.02	0.032	0.004	0.10
BMI[c]	−0.003	0.004	0.494	−0.01	0.01
Not White (ref. White)	0.02	0.05	0.693	−0.08	0.12
Random Effects					
BE Intercept	**0.49*** **	0.03	<0.001	0.45	0.53
Slope BE$_{t-1}$→ BE	**0.22*** **	0.01	<0.001	0.20	0.24
DV: Eating to Cope					
Within					
Intercept	**1.06*** **	0.25	<0.001	0.56	1.55
Vigilance $_{t-1}$[a]	0.01	0.05	0.803	−0.08	0.10
Time	**0.02***	0.01	0.023	0.00	0.03
Eating to Cope $_{t-1}$	−0.01	0.03	0.730	−0.07	0.05
Between					
EWS	0.05	0.23	0.804	−0.40	0.52
Vigilance[b]	**0.40*** **	0.07	<0.001	0.26	0.55
BMI[c]	**−0.03***	0.01	0.038	−0.06	−0.00
Not White (ref. White)	0.20	0.17	0.225	−0.13	0.53
Random Effects					
EC Intercept	**0.58*** **	0.09	<0.001	0.42	0.79
Slope time → EC	**0.002** **	0.001	0.006	0.001	0.004

Note:
[a] Person mean-centered.
[b] Aggregated.
[c] Grand mean-centered; EWS = experienced weight stigma, RD = restrictive dieting, BE = binge eating, EC = eating to cope.

Table 5
Alternative tests of binge eating.

	B	SE	p	LLCI	ULCI
Poisson Model					
Within					
Intercept	**−4.17*** **	0.67	<0.001	−5.48	−2.86
Vigilance $_{t-1}$[a]	−0.19	0.10	0.059	−0.40	0.01
Time	0.03	0.01	0.059	−0.001	0.06
Binge Eating $_{t-1}$	−0.48	0.28	0.093	−1.03	0.08
Between					
EWS	0.68	0.56	0.229	−0.42	1.77
Vigilance[b]	0.31	0.16	0.057	−0.01	0.62
BMI[c]	−0.02	0.03	0.560	−0.08	0.04
Not White (ref. White)	0.06	0.35	0.861	−0.63	0.76
Random Effects					
BE Intercept	**1.82**	0.52		1.03	3.19
Slope BE$_{t-1}$→ BE	**0.57**	0.44		0.13	2.55
Negative Binomial Model					
Within					
Intercept	**−4.07*** **	0.67	<0.001	−5.38	−2.75
Vigilance $_{t-1}$[a]	−0.21	0.13	0.109	−0.46	0.05
Time	0.03	0.02	0.136	−0.01	0.06
Binge Eating $_{t-1}$	−0.29	0.29	0.307	−0.85	0.27
Between					
EWS	0.67	0.56	0.229	−0.42	1.76
Vigilance[b]	0.31	0.16	0.052	−0.003	0.63
BMI[c]	−0.02	0.03	0.586	−0.08	0.04
Not White (ref. White)	0.03	0.35	0.929	−0.66	0.72
Random Effects					
BE Intercept	**1.55**	0.50		0.82	2.92
Slope BE$_{t-1}$→ BE	**0.37**	0.40		0.04	3.07

Note:
[a] Person mean-centered.
[b] Aggregated.
[c] Grand mean-centered; EWS = experienced weight stigma, BE = binge eating.

to hypotheses, the interaction between experienced weight stigma and internalized weight stigma did not predict vigilant coping. Within-person vigilant coping predicted restricted dieting at the next timepoint, as hypothesized, but contrary to hypotheses, it did not predict binge eating or eating to cope at the next timepoint. Between-person vigilant coping, however, predicted all three eating outcomes.

This study examined interpersonal sources of weight stigma as predictors of vigilant coping. However, throughout the week, participants' primary report of day-to-day weight stigma was structural (i.e., spaces or equipment being too small for their bodies). So, in the future, the impact of these structural sources of weight stigma should be further considered, given failure to fit properly into spaces or equipment may be a daily source of worry or expectation in day-to-day life. Additionally, the scale of weight stigma, while commonly used in the literature, has not been validated. As such, it is important that more work is done on measurement of broad weight stigma experiences beyond the frequency of specific instances (e.g., being treated as lazy), since these tend to show floor effects (Carels et al., 2019; Potter et al., 2019, 2021).

This sample was high in internalized weight stigma, $M = 4.6$, and most participants had experienced interpersonal weight stigma in their lifetime (87.8%). While this is a high frequency of participants to have experienced weight stigma in the past, compared to general samples (Puhl et al., 2021; Sonneville et al., 2024), it aligns with others (Remmert et al., 2019; Seacat et al., 2016) and is likely due to this sample being restricted to women who perceive themselves to be higher weight. Those with higher body weights typically report more weight stigma than those with average body weights (Himmelstein et al., 2017), and weight stigma increases as BMI increases in international samples among cisgender women (Puhl et al., 2021). This sample had a similar average score on the Weight Bias Internalization Scale as a typical clinical weight loss sample, who typically have around a mean of 4.5 compared to general samples with a mean of 3.5 (Romano et al., 2022). Therefore, the moderating effect may have been mitigated by the overall high mean internalization and experienced weight stigma within the sample, since these are not representative of the full range of values found in a representative sample. It should also be noted that these results may not generalize to other samples because of these unique characteristics. Additionally, BMI was uncorrelated with EWS and IWS, suggesting that it may not be a useful predictor or control of their relationships with health. However, this sample had a limited BMI range, so this may be due to the restricted range of BMI rather than due to them being uncorrelated.

Because vigilant coping predicted restrictive dieting, specific body-related mechanisms of vigilant coping and health behaviors should be explored. Previous research has focused on how weight stigma increases people's overall stress; however, these results suggest that beyond just general stress, weight-related vigilance is also an important mechanism for restrictive dieting, potentially due to motivations to lose weight to avoid stigma. However, restrictive dieting is linked to long-term weight gain rather than long-term weight loss (Neumark-Sztainer et al., 2007; Tomiyama et al., 2010). Even though restrictive dieting may contribute to short-term weight loss, over time, people typically gain more weight back (Neumark-Sztainer et al., 2007; Tomiyama et al., 2010). So, this may provide a critical link for how vigilance may contribute to the link between weight stigma and future weight gain, in line with previous

models of weight stigma and health (Tomiyama, 2014). Our participants' responses to vigilance with restriction suggest a strong motivation to attempt to escape the stigmatized identity (by trying to lose weight; Hunger et al., 2015). Therefore, understanding how escape, avoidance, and self-regulation are related to vigilant coping is warranted (Hunger et al., 2015; Wetzel and Himmelstein, 2023). Since Hunger et al., 2015, little research has examined the proposed model of weight-based social identity threat as outlined by the review, so much of the same needs highlighted in that paper are still needed today.

Future research should also investigate how vigilant coping conceptually and operationally extends to weight-related vigilant coping. Weight stigma and racism differ in key ways regarding affect and perceived injustice (Brondolo et al., 2008; Durso and Latner, 2008). Thus far, research has replicated from race-based to weight-based social identity threat in physiological responses to anticipating stigma (Hunger et al., 2018; Sawyer et al., 2012). However, continued work should be done to understand people's experiences with vigilant coping and how they may differ by identity and context. Specifically, understanding experiences of weight-related vigilance from those of other genders (e.g., men, transgender and gender non-conforming people) will be important, as well as simply replicating these findings in larger racially and gender diverse samples.

One limitation of this study was the measurement of eating outcomes. The measurements used were limited to 2–3 items to lessen attrition and participant burden; however, in the future, validated measures should be used to increase trust in these results. Additionally, in the future, items should ensure mutually exclusive responses, such as asking if participants had eaten emotionally *not including a binge episode* to ensure they are reporting exclusive events. Also, with the measures included in this study, participants may have responded "Disagree" because they didn't eat *or* "Disagree" because they ate but did not engage in maladaptive eating, which may create issues of validity of the measure. Limiting the sample to cis-gender, college-age women is a notable limitation. However, due to limited resources, the university subject pool was utilized for recruitment for this study, which has few men with higher body weight. Thus men, transgender, and non-binary individuals were not eligible due to the in-feasibility of adequate recruitment for meaningful statistical comparisons. In terms of race and ethnicity, this sample is similar to the last US Census (2020), with a slight overrepresentation of Black and White Americans. Despite having relative proportional representation, the cell sizes for race-ethnicity did not allow for meaningful statistical comparison. As described previously, it is important that this field better understands these relationships across gender groups as well as racial-ethnic groups and their intersections. Finally, this study was entirely self-report, which is a limitation due to its subjective nature, rather than some objective form of data (e.g., physiological data).

This study was strong in its design, since existing EMA research on weight stigma has been mostly descriptive or has examined only between-persons effects of weight stigma and health outcomes (Bidstrup et al., 2023). This study was pre-registered (though deviated from pre-registration), increasing trust in transparency and justifiability of design and analytic decisions. While there have been an increasing number of EMA studies in the weight stigma literature, almost none of them have included pre-registration of design or analysis (Bidstrup et al., 2023). There have also been no other peer-reviewed longitudinal studies examining weight-related vigilant coping, so this is the first study to show that weight-related vigilant coping may predict one's behaviors and cognitions (i.e., restrictive dieting). This area of research should, in the future, be expanded to understand how individuals with marginalized racial identities and other gender identities engage in vigilant coping due to weight. Specifically, Black individuals may be more at risk of vigilant coping than White participants (Wetzel and Himmelstein, 2023). Additionally, the next line in this area of research should be to experimentally examine the effects of vigilance on eating behaviors. While this study provides a naturalistic rationale of temporal precedence, we cannot control for all external factors, and thus, experimental evidence will be crucial for the understanding of the impact of vigilance on health behaviors.

8. Conclusion

Experienced and internalized weight stigma independently predict vigilant coping in everyday life but do not interact to do so among a sample of higher weight, female college students. Additionally, weight-related vigilant coping earlier in the day predicted restrictive dieting later in the day, but not emotional or binge eating. Future research should continue to investigate the relationships between these constructs to better understand weight-based social identity threat and its health consequences.

CRediT authorship contribution statement

Karen E. Wetzel: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mary S. Himmelstein:** Writing – review & editing, Visualization, Supervision, Resources, Funding acquisition, Conceptualization. **Jeffrey A. Ciesla:** Writing – review & editing, Methodology, Formal analysis.

Ethical approval

This study was conducted in line with ethical approval by Kent State University IRB (#322).

Data availability

Data and code is available via the OSF link.

References

Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. J. Roy. Stat. Soc. B 57 (1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x.

Bidstrup, H., Brennan, L., Kaufmann, L., Meadows, A., De La Piedad Garcia, X., 2023. A systematic review of ecological momentary assessment studies on weight stigma and a call for a large-scale collaboration. Obes. Rev., e13676 https://doi.org/10.1111/obr.13676.

Blodorn, A., Major, B., Hunger, J., Miller, C., 2016. Unpacking the psychological weight of weight stigma: a rejection-expectation pathway. J. Exp. Soc. Psychol. 63, 69–76. https://doi.org/10.1016/j.jesp.2015.12.003.

Bolger, N., Laurenceau, J.-P., 2013. Intensive Longitudinal Methods: an Introduction to Diary and Experience Sampling Research. The Guilford Press.

Brondolo, E., Brady, N., Thompson, S., Tobin, J.N., Cassells, A., Sweeney, M., Mcfarlane, D., Contrada, R.J., 2008. Perceived racism and negative affect: analyses of trait and state measures of affect in a community sample. J. Soc. Clin. Psychol. 27 (2), 150–173. https://doi.org/10.1521/jscp.2008.27.2.150.

Brownlow, B.N., 2023. How racism "gets under the skin": an examination of the physical- and mental-health costs of culturally compelled coping. Perspect. Psychol. Sci. 18 (3), 576–596. https://doi.org/10.1177/17456916221113762.

Brownlow, B.N., Cheavens, J.S., Vasey, M.W., Thayer, J.F., Hill, L.K., 2023. Culturally compelled coping and depressive symptoms in Black Americans: examining the role of psychophysiological regulatory capacity. Emotion. https://doi.org/10.1037/emo0001323.

Carels, R.A., Hlavka, R., Selensky, J.C., Solar, C., Rossi, J., Caroline Miller, J., 2019. A daily diary study of internalised weight bias and its psychological, eating and exercise correlates. Psychol. Health 34 (3), 306–320. https://doi.org/10.1080/08870446.2018.1525491.

Charlesworth, T.E.S., Banaji, M.R., 2019. Patterns of implicit and explicit attitudes: I. Long-term change and stability from 2007 to 2016. Psychol. Sci. 30 (2), 174–192. https://doi.org/10.1177/0956797618813087.

Clark, R., Anderson, N.B., Clark, V.R., Williams, D.R., 1999. Racism as a stressor for African Americans: a biopsychosocial model. Am. Psychol. 54 (10), 805–816.

Crocker, J., Major, B., 1989. Social stigma and self-esteem: the self-protective properties of stigma. Psychol. Rev. 96 (4), 608–630, 0033-295X/89.

Durso, L.E., Latner, J.D., 2008. Understanding self-directed stigma: development of the weight bias internalization scale. Obesity 16 (Suppl. 2), S80–S86. https://doi.org/10.1038/oby.2008.448.

Fairburn, C.G., Beglin, S.J., 1994. Eating Disorder Examination Questionnaire (EDE-Q). APA PsycTests.

Gerend, M.A., Patel, S., Ott, N., Wetzel, K., Sutin, A.R., Terracciano, A., Maner, J.K., 2021. A qualitative analysis of people's experiences with weight-based discrimination. Psychol. Health 1–18. https://doi.org/10.1080/08870446.2021.1921179.

Gerend, M.A., Zetrenne, S., Sutin, A.R., Naar, S., Maner, J.K., 2023. Weight discrimination and health risk behavior in racial, ethnic, and sexual minority adults. Ann. Behav. Med. 57 (7), 571–581. https://doi.org/10.1093/abm/kaad003.

Hales, C.M., 2020. Prevalence of Obesity and Severe Obesity Among Adults: United States, 2017–2018, p. 360.

Hicken, M.T., Lee, H., Hing, A.K., 2018. The weight of racism: vigilance and racial inequalities in weight-related measures. Soc. Sci. Med. 199, 157–166. https://doi.org/10.1016/j.socscimed.2017.03.058.

Hunger, J.M., Major, B., Blodorn, A., Miller, C.T., 2015. Weighed down by stigma: how weight-based social identity threat contributes to weight gain and poor health. Soc. Pers. Psychol. Compass 9 (6), 255–268. https://doi.org/10.1111/spc3.12172.

Himmelstein, M.S., Incollingo Belsky, Tomiyama, A.J., 2015. The weight of stigma: cortisol reactivity to manipulated weight stigma: the Weight of Stigma. Obesity 23 (2), 368–374. https://doi.org/10.1002/oby.20959.

Himmelstein, M.S., Puhl, R.M., Quinn, D.M., 2017. Intersectionality: an understudied framework for addressing Weight Stigma. Am. J. Prev. Med. 53 (4), 421–431. https://doi.org/10.1016/j.amepre.2017.04.003.

Himmelstein, M.S., Tomiyama, A.J., 2015. It's not you, it's me: self-perceptions, antifat attitudes, and stereotyping of obese individuals. Soc. Psychol. Personal. Sci. 6 (7), 749–757. https://doi.org/10.1177/1948550615585831.

Himmelstein, M.S., Young, D.M., Sanchez, D.T., Jackson, J.S., 2015. Vigilance in the discrimination-stress model for Black Americans. Psychol. Health 30 (3), 253–267. https://doi.org/10.1080/08870446.2014.966104.

Hunger, J.M., Blodorn, A., Miller, C.T., Major, B., 2018. The psychological and physiological effects of interacting with an anti-fat peer. Body Image 27, 148–155. https://doi.org/10.1016/j.bodyim.2018.09.002.

Hunger, J.M., Dodd, D.R., Smith, A.R., 2020. Weight discrimination, anticipated weight stigma, and disordered eating. Eat. Behav. 37, 101383. https://doi.org/10.1016/j.eatbeh.2020.101383.

Jackson, J., Williams, D., 2002. Detroit area study, 1995: social influence on health: stress, racism, and health protective resources, Inter-University Consortium for Political and Social Research.

Jackson, B., Cooper, M.L., Mintz, L., Albino, A., 2003. Motivations to eat: scale development and validation. J. Res. Pers. 37 (4), 297–318. https://doi.org/10.1016/S0092-6566(02)00574-3.

Levinson, J.A., Kinkel-Ram, S., Myers, B., Hunger, J.M., 2024. A systematic review of weight stigma and disordered eating cognitions and behaviors. Body Image 48, 101678. https://doi.org/10.1016/j.bodyim.2023.101678.

Major, B., O'Brien, L.T., 2005. The social psychology of stigma. Annu. Rev. Psychol. 56, 393–421. https://doi.org/10.1146/annurev.psych.56.091103.070137.

Major, B., Hunger, J.M., Bunyan, D.P., Miller, C.T., 2014. The ironic effects of weight stigma. J. Exp. Soc. Psychol. 51, 74–80. https://doi.org/10.1016/j.jesp.2013.11.009.

Major, B., Schmader, T., 2018. Stigma, social identity threat, and health. In: Major, B., Dovidio, J.F., Link, B.G. (Eds.), *The Oxford handbook of stigma, discrimination, and health*. Oxford University Press, pp. 85–103.

Neumark-Sztainer, D., Wall, M., Haines, J., Story, M., Eisenberg, M.E., 2007. Why does dieting predict weight gain in adolescents? Findings from project EAT-II: a 5-year longitudinal study. J. Am. Diet. Assoc. 107 (3), 448–455. https://doi.org/10.1016/j.jada.2006.12.013.

Pearl, R.L., Puhl, R.M., 2014. Measuring internalized weight attitudes across body weight categories: validation of the modified weight bias internalization scale. Body Image 11 (1), 89–92. https://doi.org/10.1016/j.bodyim.2013.09.005.

Pearl, R.L., Wadden, T.A., Jakicic, J.M., 2021. Is weight stigma associated with physical activity? A systematic review. Obesity 29, 1994–2012. https://doi.org/10.1002/oby.23274.

Pearl, R.L., Wadden, T.A., Bach, C., LaFata, E.M., Gautam, S., Leonard, S., Berkowitz, R. I., Latner, J.D., Jakicic, J.M., 2023. Long-term effects of an internalized weight stigma intervention: a randomized controlled trial. J. Consult. Clin. Psychol. 91 (7), 398–410. https://doi.org/10.1037/ccp0000819.

Potter, L.N., Brondolo, E., Smyth, J.M., 2019. Biopsychosocial correlates of discrimination in daily life: a review. Stigma Health 4 (1), 38–61. https://doi.org/10.1037/sah0000120.

Potter, L.N., Meadows, A., Smyth, J., 2021. Experiences of weight stigma in everyday life: an ecological momentary assessment study. J. Health Psychol. 26 (14), 2781–2793. https://doi.org/10.1177/1359105320934179.

Preacher, K.J., Zyphur, M.J., Zhang, Z., 2010. A general multilevel SEM framework for assessing multilevel mediation. Psychol. Methods 15 (3), 209–233. https://doi.org/10.1037/a0020141.

Pudney, E.V., Himmelstein, M.S., Puhl, R.M., Foster, G.D., 2020. Distressed or not distressed? A mixed methods examination of reactions to weight stigma and implications for emotional wellbeing and internalized weight bias. Soc. Sci. Med. 249. https://doi.org/10.1016/j.socscimed.2020.112854.

Puhl, R.M., Heuer, C., Sarda, V., 2011. Framing messages about weight discrimination: impact on public support for legislation. Int. J. Obes. 35, 863–872. https://doi.org/10.1038/ijo.2010.194.

Puhl, R.M., Himmelstein, M.S., Pearl, R.L., 2020. Weight stigma as a psychosocial contributor to obesity. Am. Psychol. 75 (2), 274–289. https://doi.org/10.1037/amp0000538.

Puhl, R.M., Lessard, L.M., Pearl, R.L., Himmelstein, M.S., Foster, G.D., 2021. International comparisons of weight stigma: addressing a void in the field. Int. J. Obes. 45 (9), 1976–1985. https://doi.org/10.1038/s41366-021-00860-z.

Remmert, J.E., Convertino, A.D., Roberts, S.R., Godfrey, K.M., Butryn, M.L., 2019. Stigmatizing weight experiences in health care: associations with BMI and eating behaviours. Obesity Sci. Prac. 5 (6), 555–563. https://doi.org/10.1002/osp4.379.

Robinson, W.S., 1950. Ecological correlations and the behavior of individuals. Am. Socio. Rev. 15 (3), 351–357.

Robinson, E., Hunger, J.M., Daly, M., 2015. Perceived weight status and risk of weight gain across life in US and UK adults. Int. J. Obes. 39 (12), 1721–1726. https://doi.org/10.1038/ijo.2015.143.

Rockwood, N.J., 2017. In: Advancing the formulation and testing of multilevel mediation and moderated mediation models. [The Ohio State Unviersity]. https://etd.ohiolink.edu/acprod/odb_etd/ws/send_file/send?accession=osu1489578419777238&dispositions=inline.

Romano, K.A., Heron, K.E., Sandoval, C.M., Howard, L.M., MacIntyre, R.I., Mason, T.B., 2022. A meta-analysis of associations between weight bias internalization and conceptually-related correlates: a step towards improving construct validity. Clin. Psychol. Rev. 92, 102127. https://doi.org/10.1016/j.cpr.2022.102127.

Sawyer, P.J., Major, B., Casad, B.J., Townsend, S.S.M., Mendes, W.B., 2012. Discrimination and the stress response: psychological and physiological consequences of anticipating prejudice in interethnic interactions. Am. J. Publ. Health 102 (5), 1020–1026. https://doi.org/10.2105/AJPH.2011.300620.

Seacat, J.D., Dougal, S.C., Roy, D., 2016. A daily diary assessment of female weight stigmatization. J. Health Psychol. 21 (2), 228–240. https://doi.org/10.1177/1359105314525067.

Selvin, H.C., 1958. Durkheim's suicide and problems of empirical research. Am. J. Sociol. 63 (6), 607–619. https://doi.org/10.1086/222356.

Sewell, W., Horsford, C.E., Coleman, K., Watkins, C.S., 2016. Vile vigilance: an integrated theoretical framework for understanding the state of Black surveillance. J. Hum. Behav. Soc. Environ. 26 (3–4), 287–302. https://doi.org/10.1080/10911359.2015.1127735.

Smyth, J., Wonderlich, S., Crosby, R., Miltenberger, R., Mitchell, J., Rorty, M., 2001. The use of ecological momentary assessment approaches in eating disorder research. Int. J. Eat. Disord. 30 (1), 83–95.

Sonneville, K.R., Thurston, I.B., Gordon, A.R., Richmond, T.K., Weeks, H.M., Lipson, S.K., 2024. Weight stigma associated with mental health concerns among college students. Am. J. Prev. Med. 66 (2), 260–268. https://doi.org/10.1016/j.amepre.2023.09.021.

Spencer, S.J., Steele, C.M., Quinn, D.M., 1999. Stereotype threat and women's math performance. J. Exp. Soc. Psychol. 35 (1), 4–28. https://doi.org/10.1006/jesp.1998.1373.

Steele, C.M., Aronson, J., 1995. Stereotype threat and the intellectual test performance of African Americans. J. Pers. Soc. Psychol. 69 (5), 797–811.

Steele, C.M., Spencer, S.J., Aronson, J., 2002. Contending with group image: the psychology of stereotype and social identity threat. Adv. Exp. Soc. Psychol. 34, 379–440. https://doi.org/10.1016/S0065-2601(02)80009-0.

Sutin, A.R., Terracciano, A., 2017. Sources of weight discrimination and health. Stigma Health 2 (1), 23–27. https://doi.org/10.1037/sah0000037.

Tajfel, H., 1970. Experiments in Intergroup Discrimination. Sci. Am. 223 (5), 96–103.

Tajfel, H., Billig, M.G., Bundy, R.P., Flament, C., 1971. Social categorization and intergroup behaviour. Eur. J. Soc. Psychol. 1 (2), 149–178. https://doi.org/10.1002/ejsp.2420010202.

Tomiyama, A.J., 2014. Weight stigma is stressful. A review of evidence for the cyclic Obesity/weight-based stigma model. Appetite 82, 8–15. https://doi.org/10.1016/j.appet.2014.06.108.

Tomiyama, A.J., Mann, T., Vinas, D., Hunger, J.M., DeJager, J., Taylor, S.E., 2010. Low calorie dieting increases cortisol. Psychosom. Med. 72 (4), 357–364.

Vartanian, L.R., Porter, A.M., 2016. Weight stigma and eating behavior: a review of the literature. Appetite 102, 3–14. https://doi.org/10.1016/j.appet.2016.01.034.

Wellman, J.D., Araiza, A.M., Solano, C., Berru, E., 2019. Sex differences in the relationships among weight stigma, depression, and binge eating. Appetite 133, 166–173. https://doi.org/10.1016/j.appet.2018.10.029.

Wetzel, K.E., Himmelstein, M.S., 2023. Constant vigilance: The impact of weight stigma, vigilance, and internalization on maladaptive eating behaviors. Health Psychol. https://doi.org/10.1037/hea0001324.

Wetzel, K.E., Himmelstein, M.S., 2023. Health care avoidance as vigilance: A model of maladaptive eating behaviors due to weight stigma in health care, avoidance, and internalization among women. Stigma and Health. https://doi.org/10.1037/sah0000470.

Wu, Y.K., Berry, D.C., 2018. Impact of weight stigma on physiological and psychological health outcomes for overweight and obese adults: a systematic review. J. Adv. Nurs. 74, 1030–1042. https://doi.org/10.1111/jan.13511.

Zhang, Z., Zyphur, M.J., Preacher, K.J., 2009. Testing multilevel mediation using hierarchical linear models: problems and solutions. Organ. Res. Methods 12 (4), 695–719. https://doi.org/10.1177/1094428108327450.